

School of Science, Computing and Engineering Technologies
Higher Education Division

Assessment Components and Assessment Criteria

COS30018

Intelligent Systems

Semester 2, 2025



Assessment Components and Assessment Criteria

Overview

Results in this unit are determined from two assessment components:

1. Project work (50%)
2. Continuous Oral Defense (50%)

Project work (100 marks):

For this assessment component (**PROJ**), you can choose one of two options:

Option A:

You are given a project topic to develop a solution for the Vehicle Routing Problem (VRP), a combinatorial optimization problem, and you will form a team of 3-4 students to work on the project. Your tutor will monitor your project throughout the semester to ensure that you make steady progress. You will deliver the complete project at the end of the semester.

Option B:

You are given a project topic to develop a solution for the Handwritten Number Recognition Problem (HNRP), a machine learning problem, and you will form a team of 3-4 students to work on the project. Your tutor will monitor your project throughout the semester to ensure that you make steady progress. You will deliver the complete project at the end of the semester.

Option C:

Some students may not be sufficiently confident in their programming abilities and can choose **Option C** after getting approval by the Convenor (bvo@swin.edu.au). For **Option C**, you are going to work in a project led by a teaching staff (i.e., the *project leader*). The project leader will provide you with an initial code base for this project. It is quite poor (in terms of programming and performance). We will give you weekly tasks to do to improve this code base (and in the process, you will learn about things such as data processing and deep learning).

Option D:

Many students may have heard about large language models (LLMs) and how they can be used to power autonomous agents (called LLM agents). **Option D** gives students an opportunity to explore this topic. This option typically requires students to put in extra work and the students must be quite capable especially in Python programming: *To do Option D, you will need to apply with the unit convenor (bvo@swin.edu.au) by presenting your suitability to do this research-oriented project: Your current average from your academic transcript has to be at least D ($\geq 70\%$) AND you are aiming to achieve D/HD in Intelligent Systems AND you are well equipped to do this project by having a strong capability in Python programming.*

We are planning to have no more than three teams for Option D. Thus, even if you satisfy all of the above requirements, but if you apply to do Option D too late and all Option D-teams have been full then you may have to choose one of the Options A, B, or C.

Continuous Oral Defence (100 marks):

For this assessment component (**COD**), you have to attend at least two (2) oral defense sessions with a teaching staff (your assessor) to answer some questions about the unit content and, possibly, also one question at the end of the semester about the project you participated in. The assessor will provide you with an arrangement of when you are having these COD sessions.

You will receive immediate feedback on your performance on these questions. In the two sessions 1 and 2, you will be asked a total of 5 (five) questions. Each question will give you a maximum of 15 marks. Thus, **after completing Sessions 1 and 2, you will get a maximum of 75 marks.**

Some students may be eligible for Session 3:

1. If the total mark you receive from Sessions 1 & 2 is less than 50, then you are eligible to attend Session 3 to improve your COD total, but your final COD total cannot be greater than 50.
2. If the total mark you receive from Sessions 1 & 2 is at least 60, then you are eligible to attend Session 3 to achieve high marks (**76-100**) and the questions you will be asked will be quite advanced to allow you to demonstrate that you have an outstanding grasp of the subject matter.

If you are eligible for Session 3, you will need to register with your assessor to attend Session 3. Session 3 is to take place in Weeks 12 and 13. The questions asked during Session 3 can be related to any week of the semester (1-11) or even about the project you are involved in.